MACROECONOMICS
ECN 1215

GOVERNMENT SECTOR

Dale Mudenda @2023

Outline

1. Aggregate demand and Aggregate Supply
   a. Aggregate Demand
   b. Aggregate supply
   c. Equilibrium
2. Fiscal Policy
   a. Governments spending and taxation
   b. Budget deficits
   c. Government debt

What is aggregate demand

- The term aggregate demand describe the total demand for all goods and services in the economy.
- It means adding demand across all individual markets for goods and services
  - We did this under measuring GDP   which adds up all the goods and services in the market by translating them into a common unit: market value
- AD measures the total quantity of goods and services demanded in the economy in terms of their market value.
- The AD curve shows he relationship between the overall price level and the level of total demand in the economy. The price is on the vertical axis and output (Y) is on the horizontal axis
  - We can use the aggregate expenditure equilibrium model to trace out AD curve and understand its shape – which is downward sloping

Aggregate Demand

- Recall, we learnt that aggregate expenditure or planned expenditure has many components that comprise- consumption, investment, government spending and net exports
  - $Ep = C + I + G + X - M$
  - The components of planned expenditure are affected by important economic variables
  - We will focus on how each component reacts to changes in the overall price level in the economy
  - We examine how real variables change in response to price changes

1. *Price and consumption* -
   - In general, changes in the price level will change the real value of people's wealth and income
   - A rise in overall price level means a given number of kwachas won't buy as much in terms of real goods and services
   - The increase in the prices reduce people's wealth in terms of their purchasing power
   - With less wealth. People reduce consumption creating a negative relationship between overall price level and consumption spending
   - This is the **wealth effect** that give a **negative** relationship between **AD** and price

Aggregate Demand

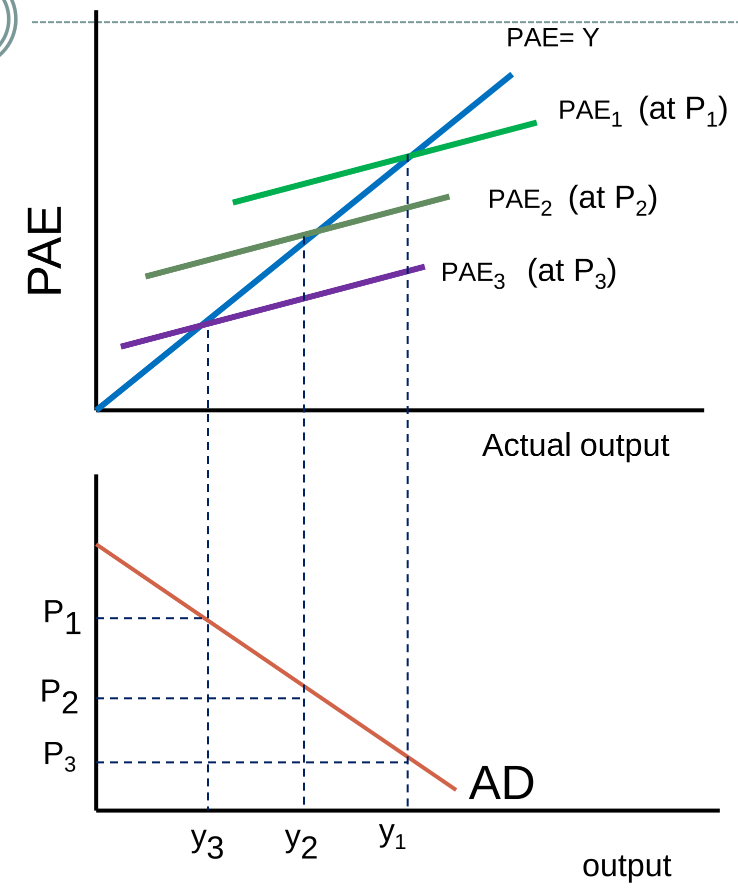
- Caveat – if wages increased as much as price increase the purchasing power of those wages will stay the same
- In this case, the wealth effect doesn't come in effect

2. *Investment and Price* – when price increases, the interest rate, that is the price of borrowing tends to increase
   - higher interest rate make it more expensive for firms to borrow and invest
   - So there is less investment in factories and working capital
   - There is an indirect negative relationship between I and prices

3. Government spending and Price
   - Government spending is mainly independent of price levels. Even when prices rise and fall, the government still needs to spend roughly the same amount to spend. So G doesn't contribute to the downward sloping of AD curve
   - Net exports -there is a negative relationship between net exports and the price level.
   - If domestic prices rise relatively to other countries, the exports imports rise while exports decrease

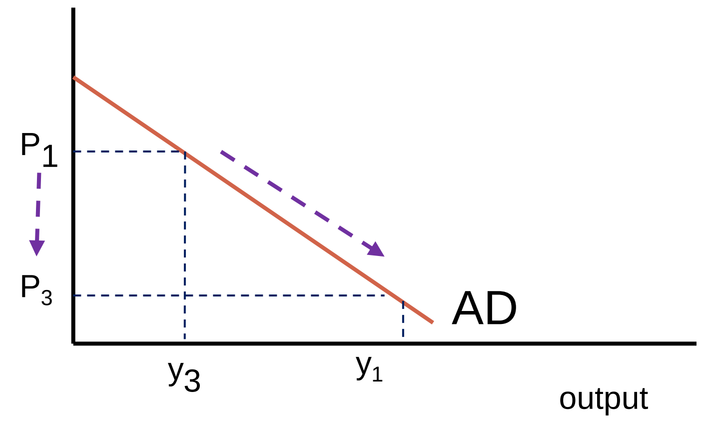
Aggregate Demand

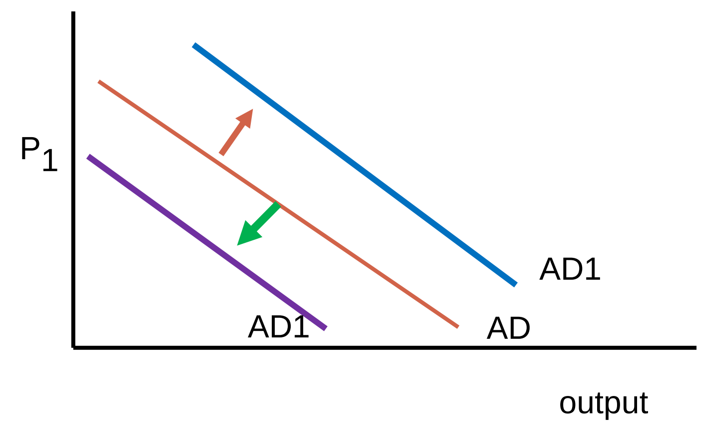
- In the figure p1 is the lowest price so p2 represents an increase from p1 and p3 represent an increase from P2
- Each price increase represent a downward shift in the planned aggregate expenditure line
- We plot the (Y, P)npairs hat correspond to the new equilibrium in the PAE or GDP
- The relationship shows that a prices decrease, the quantity demanded increases
  - One implication- anything other than a change in the price level itself that leads to a different equilibrium in aggregate expenditure model will cause the AD curve to shift



Aggregate Demand

- Shifting AD curve
  - Price changes generate movements along the AD curve
  - The entire AD curve shifts in response to non-price changes in an of the four components of AD
  - If people have more income and feel confident about the future, the are likely to consume more and the AD shifts to the right from AD to AD 1
  - If they feel less confident and consumption reduces, the AD shifts to AD3





Aggregate Demand

Category	Increase (Shift right)	Decrease (shift left)
Consumption	○ High expectations about future income increase consumer spending ○ Tax cuts increase consumer spending	○ Low expectations about future income leads to greater savings and less consumer spending ○ Higher interest rates discourage borrowing
Investment	○ Confidence in future of the economy leads to firms to expand businesses ○ Tax credit for small business inspires firms to buy new company cars	○ Firms cut back on spending in order to weather a recession ○ Taxes on capital increase, leaving less money for investment
Government spending	○ Increase in government spending spur spending after recession	○ decrease in government spending in response to concerns about increasing debt leads to less spending
Net exports	○ New free trade agreement reduces most tariffs ○ Economic growth in SA for example	Other countries increase barriers against our exports Kwacha strengthens, making Zambian goods more expensive for international consumers

Aggregate SUPPLY

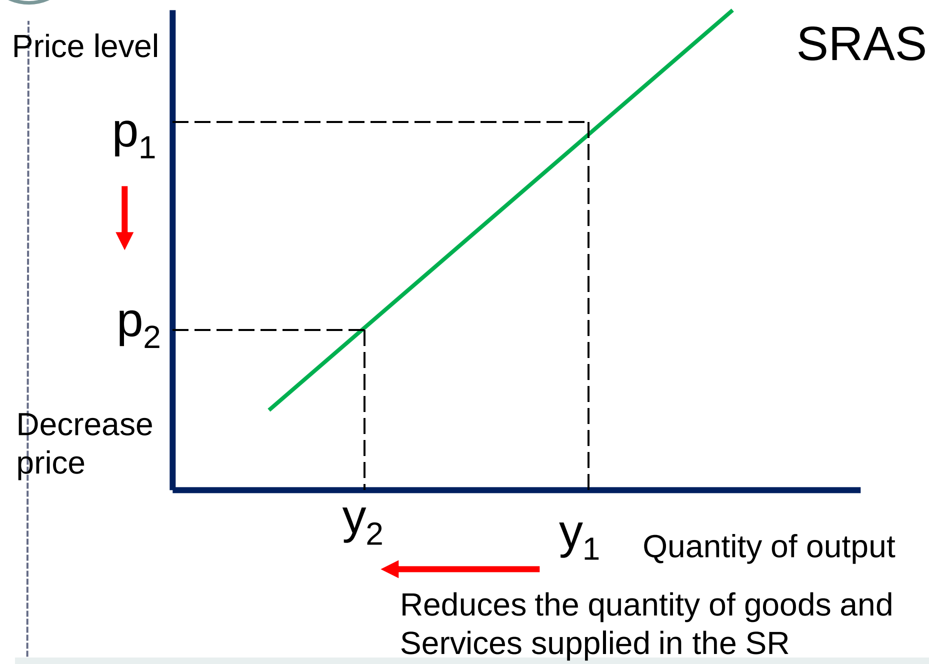
- AS – is the sum total of the production of all firms in the economy
- AS –curve shows the relationship between overall price level in the economy and total production by firms
- At the macroeconomic level there is a difference the way the economy operates in the short run and how the economy operates in the long run. As a result, the shape of the AS depends on the type horizon

1. Short-run AS

- The short run refer to hourly, daily decisions that firms make
- Some inputs are variable while others cannot be varied
  - e.g. Labour vs Capital
- In short run, the aggregate supply curve is upward sloping
- Meaning –as the overall price level increases, input prices don't all increase immediately
- As the overall price increase, there is a tendency for the quantity of goods and services supplied to rise
- A decrease in the level of prices tends to reduce the quantity of goods and services supplied

Aggregate Supply

- This shape of the AS curve emanates from the Keynesian views
- Three theories have been proposed for the upward sloping AS
  - They commonly hold that – the quantity of output supplied deviates from its long run or natural level when the price level deviates from the price level that people expected to prevail



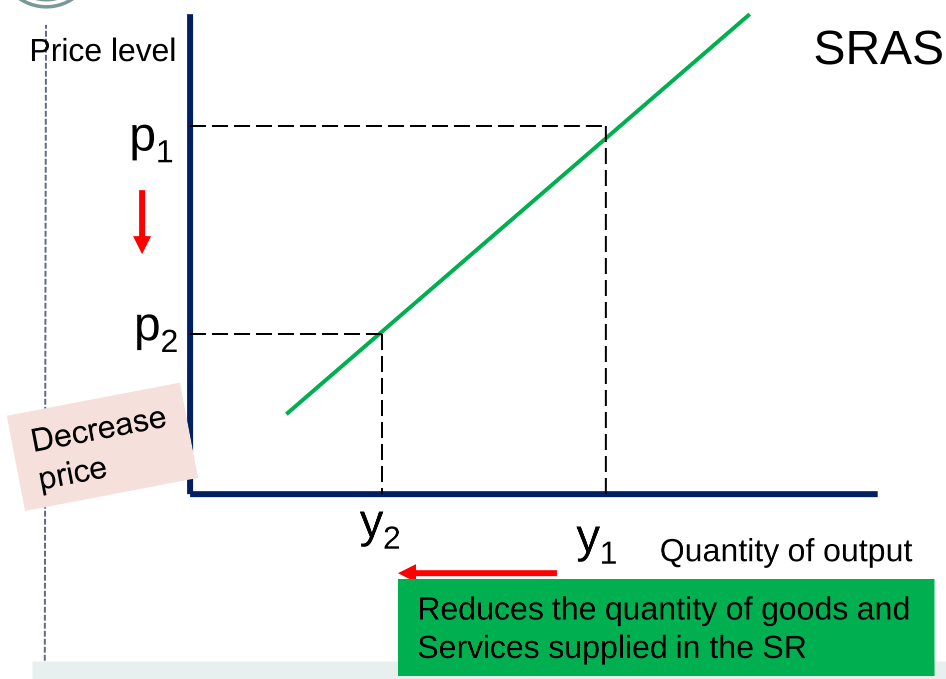
- When he price level rises above the expected level, output rises above the natural level, output rises above the natural rate and when the price level falls below the expected level, output falls below its natural rate

Aggregate Supply

- Explanations:

a) The sticky wage theory - the theory proposes that the SRAS slopes upwards because nominal wages are slow to adjust or are "sticky" in the short-run

- Wages can be slow to adjust because of contracts between workers and firms often through unions
- Contracts can be long term and fixed which prevents wages from changing in the SR
- In addition, the slow adjustment may be attributable to social norms and notions of fairness



- That influence wage setting and change only slowly over time
- Example: assume the firm and workers agree a nominal wage of W based on expected price level to be

Aggregate Supply

- If the price P falls below the level that was expected and the nominal wage is stuck at W, then the real wage **W/P rises** above he level the firm planned to pay
- Because wages are large part of the firm's production costs, a higher real wage means that the firm's real costs have risen
- The firm will respond by hiring less workers and producing less quantity

b) Sticky price theory -emphasizes that prices of some goods and services also adjust sluggishly in response to changing economic conditions

- The slow adjustment of prices occurs in part due to menu costs to adjusting prices
  - Menu costs – include cost of printing and distributing price lists
  .

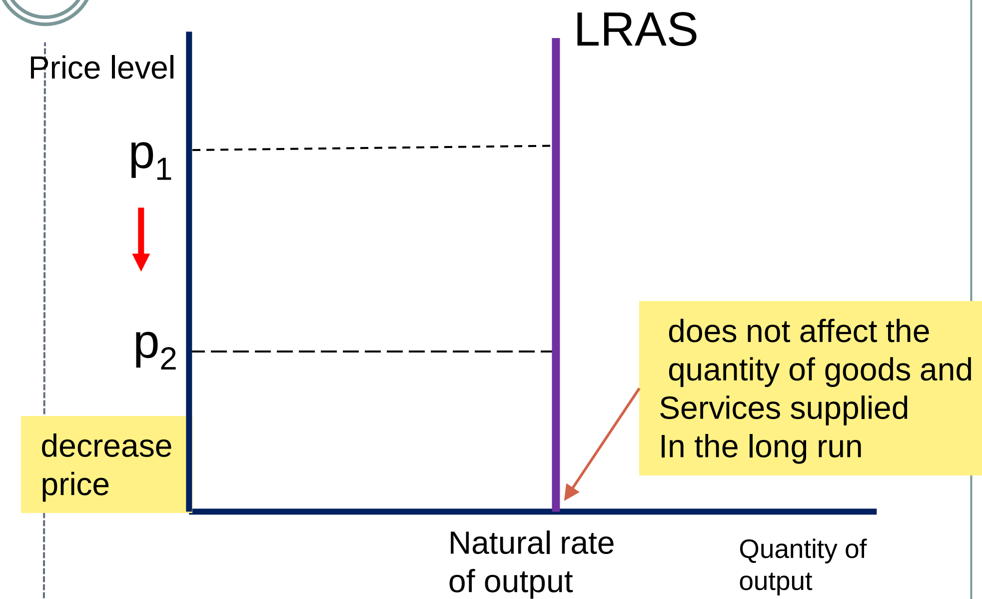
c) Misperceptions Theory - the theory assert that changes in the overall price level can temporarily mislead suppliers about what is happening in the individual markets in which they sell their products.

- For example: if the price falls below what people expected,

Aggregate Supply

- Suppliers may mistakenly believe that their relative prices have fallen
- Thus, they may reduce the supply of the product
- In this case the low price misconception about relative prices induces suppliers to respond to the lower price level by decreasing the quantity of services and goods supplied

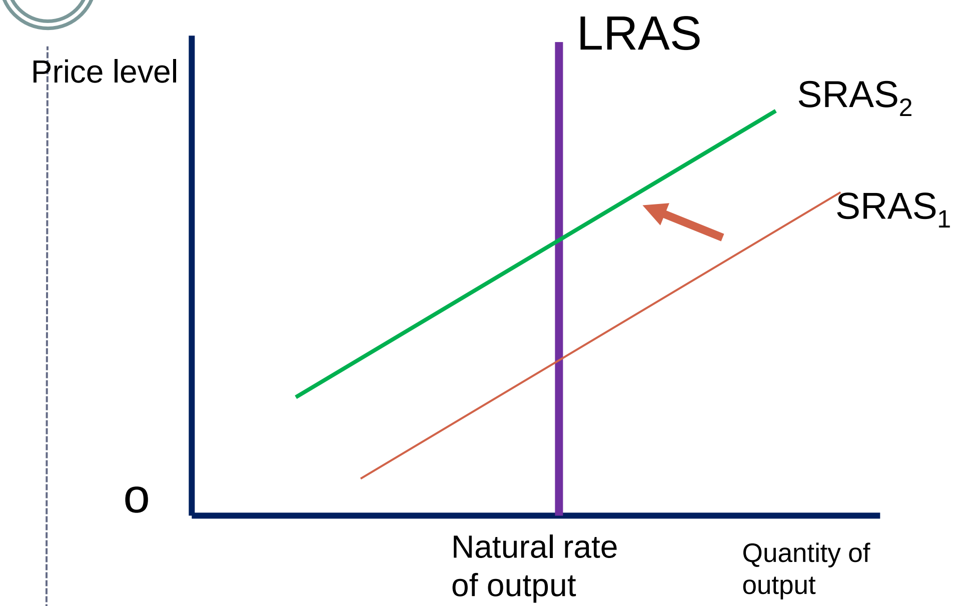
2. **LONG RUN SUPPLY CURVE**
   - In the Long-run an economy's production of goods and services depends on its supplies



- of labor, capital and natural resources + technology
- The price doesn't affect the long-term determinants of real GDP , the LRAS is upward sloping
- In line with the classical theory,

Aggregate Supply

- The LRAS represents potential output in the economy
  - i.e., the level of output possible if the economy is operating at full capacity
  - Changes in LRAS curve occur due to changes in the society's resources
- The economy does not always produce its full potential output. Sometimes less output and sometimes more output is produced
  - When output is higher than potential output, the economy is in a boom
  - When output is below potential, the economy is in a recession
- Business Cycles: fluctuations of output around the level of potential output in the economy



Shifts in the SRAS curve:

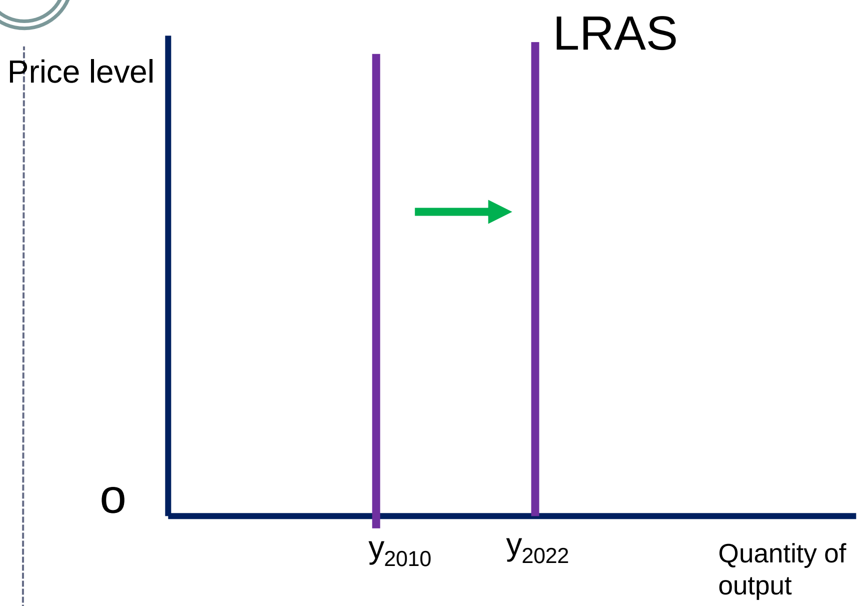
- SR AS van shift if:
- Costs of production change. If the costs of production rise, firms find it expensive to produce. The SRAD shifts to the left

Aggregate Supply

- The SRAS will also shift if there are major events also called supply shocks (e.g., flood)
- Expectations about the future prices will also shift the SRAS

- SHIFTS IN LRAS –CURVE

- In the long run, firms produce an amount dictated by available inputs regardless of the price level
  - The LRAS –curve shifts to the right if the potential output of the economy expands
  - LRAS shifts to the left if the economy loses productive capacity



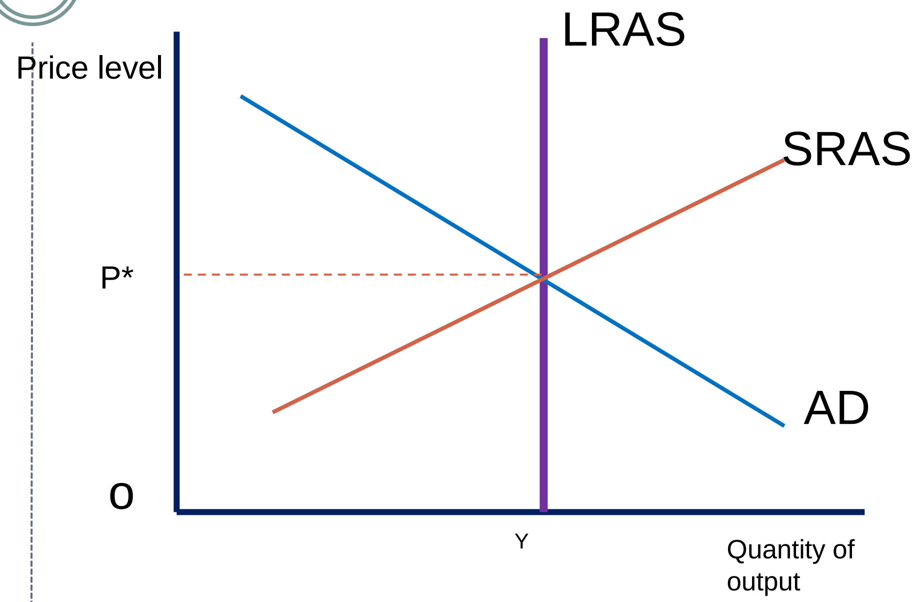
- If for example we have an increase in foreign direct investment, the productive capacity of the economy increases
- This will shift the LRAS curve to the right

Aggregate Demand

Factor	Increase LRAS	Decrease LRAS
Technology	○ Technological innovation allows for greater production using the same amount of inputs	○ A new law stripping away intellectual property rights reduces the incentive innovate
Capital	○ Foreign direct investment in factories and machines	○ Depreciation and wear break down capital
Labour	○ Immigration increases the available supply of labour	○ Aging population takes workers out of the labour force
Natural resources	○ New energy resources allow factories to produce more with the same inputs	○ Climate change permanently reduce=s the amount of land that can be farmed

Equilibrium

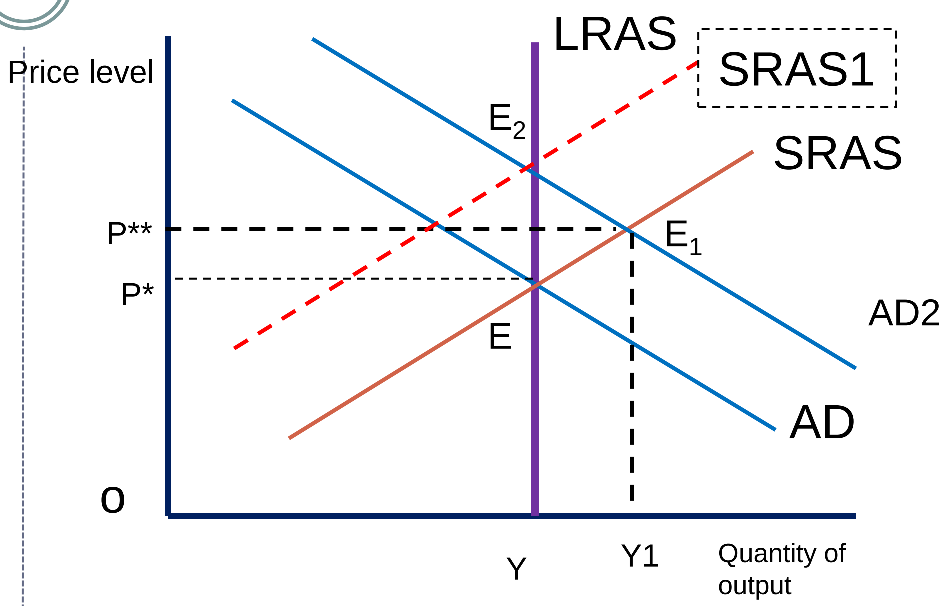
- We have the facets on the full model
  - Aggregate demand curve
  - SRAS curve
  - LRAS curve
- Equilibrium in national income is at the point at which AD is equal to AS
- Short-run equilibrium – is given by intersection of AD and SRAS curve
- In LR equilibrium – the AD crosses the long run aggregate supply curve at the same point



- Many shocks can shift the equilibrium
- Imagine there is an increase in consumption, in the SR, the AD will shift to the right

Equilibrium

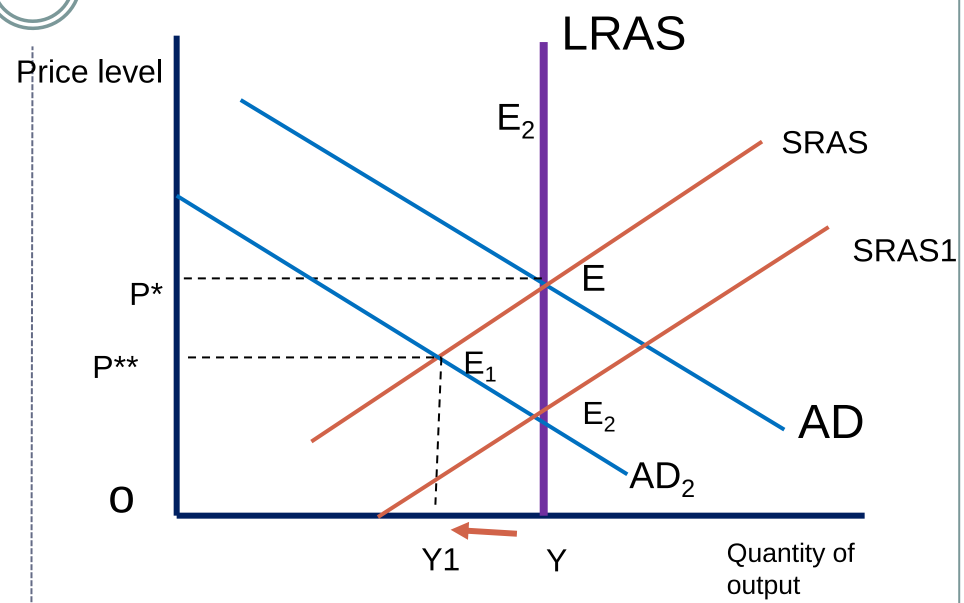
- New SR equilibrium is set at E_1 from E
- Point E_1 shows the effect of increased consumption. Output goes above the long run potential level and prices increased
- Point E_1 is an equilibrium point – it's a SR equilibrium
  - How does the economy return to the LR equilibrium?
  - The SRAS has to shift again
  - At E1, the prices rise from p* to p**
  - This causes an increase in input costs for the firms and therefore AS shifts
  - As contracts are negotiated and wages increased, the SRAS would shift to the left – at SRAS1



- LR equilibrium is attained at E2
- The change in equilibrium from E to E2 represents a movement from an initial LR equilibrium to a new LR equilibrium
- The opposite can happen –check in tutorials

Equilibrium

- Opposite – assume a demand shock that shifts the AD to the left – from AD to AD2
- This exerts pressure on the price, and they decline from p* to p**
- Inputs become cheaper and the
- New equilibrium at E1 which is the SR equilibrium
- At E1 both prices & output are lower
- As a response to the cheaper or lower inputs prices, firms produce more and AS shifts to the right to SRAS1
- The LR equilibrium moves from E to E2
  - Which restores the LR equilibrium output with the same original level of output Y but lower prices



- Practice the impact of various shocks

Effects of shift in AD

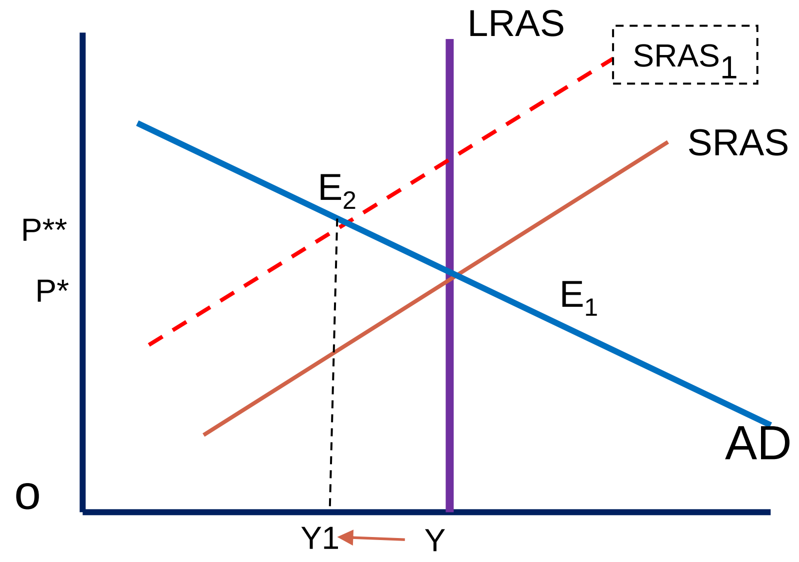
Shift	Example	Short run	Long Run
Increase in AD	Increase in G or reduction in taxes	Output increases, price increases	Output unchanged Price increases
Decrease in AD	Reduction in consumer confidence, reduces C	Output decreases price decreases	Output unchanged Price decreases

Supply Shocks

- Economies can also face supply shocks which can be:
  1. Temporary changes – result in the shift of the SRAS curve
  2. Permanent Shocks – result in both LRAS and SRAS shift

1. Temporary Shocks

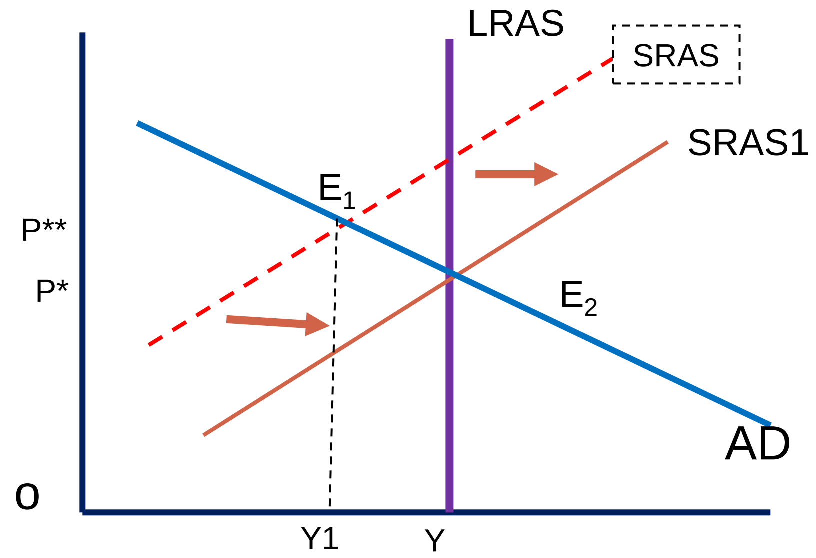
- Suppose we have a drought, and a lot of maize is damaged
- The shock shifts the supply curve to the left from SRAS to SRAS1
- The economy moves from LR equilibrium E1 to the SR equilibrium E2
- Prices are higher and output lower



- How does the economy adjust back to equilibrium:
- When output decreases while prices increase, the situation is called stagflation- that is economic stagnation coupled with high prices

Supply Shocks

- In this situation adjustment to long run equilibrium is not easy
- Drop in wages would help. However, wages tend to be inflexible downward
- This contributes to keeping the economy in this undesirable equilibrium
- When the economy is operating at less than full potential output and wages are slow to fall, firms will lay off workers
- Unemployment will be higher than its natural rate
- With high unemployment wages will fall

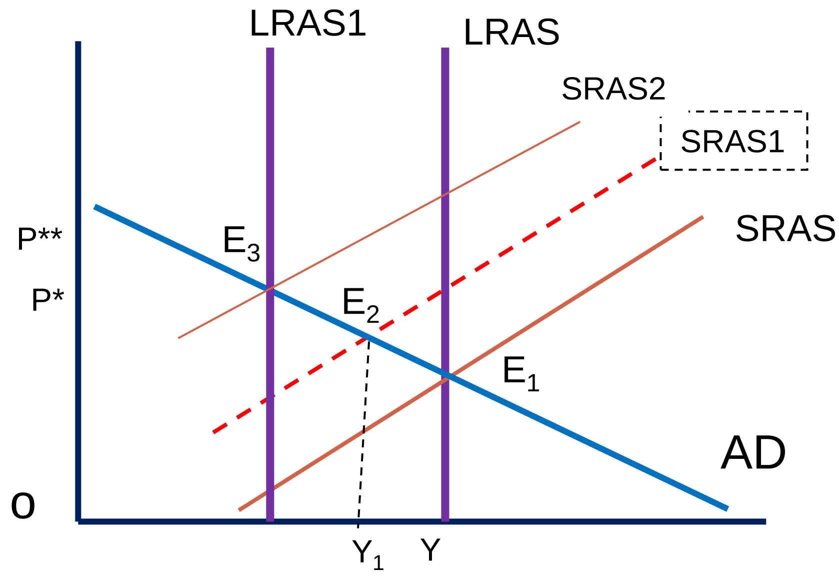


- Cheaper labour will then lower the cost of production
- The SRAS curve shifts back to the right until it returns to the old LR equilibrium E2

Supply Shocks

2. Permanent Supply Shocks

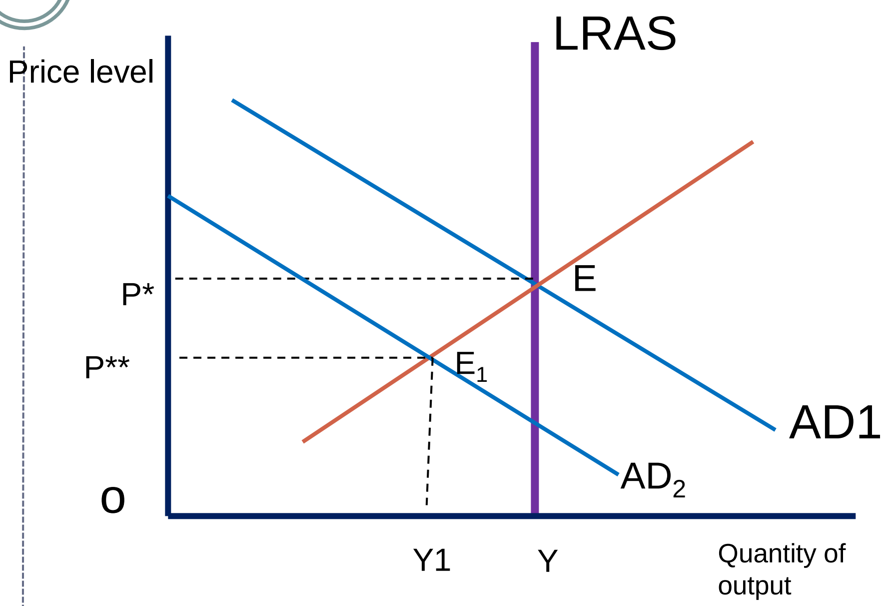
- Imagine instead of a one-year drought, a cataclysmic climate change make it impossible to grow maize for longer periods, land becomes unproductive
- The effect will have both SR and LR
- The loss of productive land will shift the LRAS curve to the left from LRAS to LRAS1
- In the short run, the resulting increase in food & other prices increases the cost of production
- The SRAS shifts to the left and new equilibrium is set at E2
- As long as the prices remain high, the SRAS will continue moving until it gets to E3



- The LR equilibrium with higher prices and lower level of potential output

Role of Public Policy

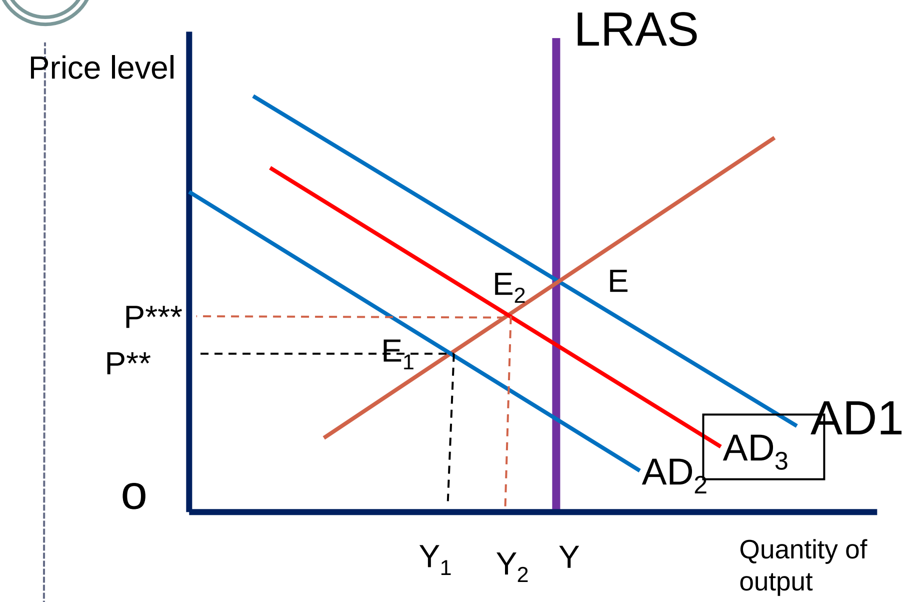
- It can take long for the economy to adjust o demand and supply shocks
- Waiting for the economy adjust itself is not easy because it hurts people due to changing prices and unemployment
- Government can try to counter the negative demand shocks by:
  - increasing government spending
  - Reducing taxes
- Increasing G shifts the AD curve to the right
- Goal –counter the negative shock with a positive one



- Assume a negative shock, such as increased taxes that reduce consumer spending is implemented
- AD declines to AD2 from AD1
- Government may implement an increase in spending

Role of Public Policy

- The increase in spending could shift the AD to AD3
- In some cases, government spending can increase AD to AD1 which restores the economy to full employment



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